Hough Transform, Two-view Geometry

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HW1

- Questions?
- Team assignment for final project
  - Your own groups, survey out after spring break
  - We will help team people up then

Recap: Procrustes

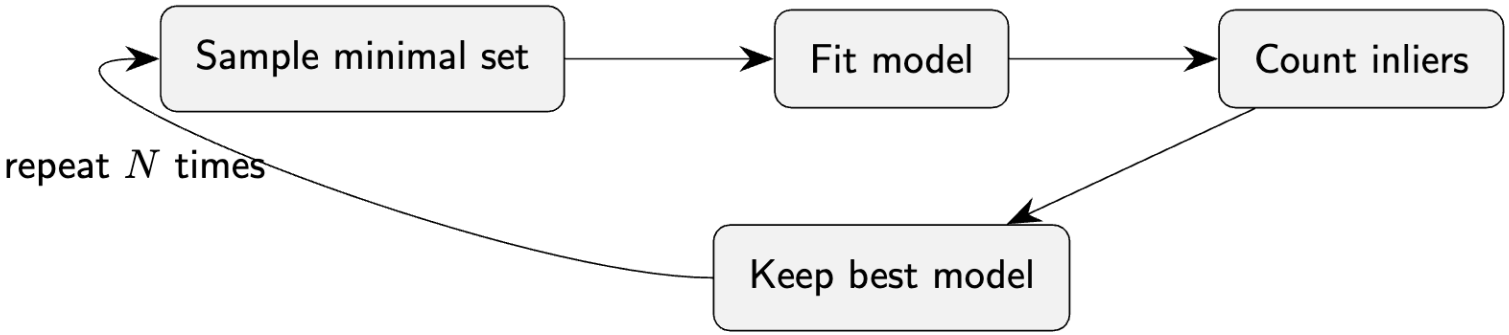
Input

Correspondences $\{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}$.

Steps

1. Compute means $\bar{\boldsymbol{x}}, \bar{\boldsymbol{y}}$ and centered matrices $\mathbf{X}_c, \mathbf{Y}_c$.
2. Compute $\mathbf{H} = \mathbf{X}_c \mathbf{Y}_c^\top$.
3. SVD: $\mathbf{H} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$.
4. Rotation: $\mathbf{R}^* = \mathbf{V}\mathbf{D}\mathbf{U}^\top$ (with reflection fix).
5. Translation: $\mathbf{t}^* = \bar{\boldsymbol{y}} - \mathbf{R}^* \bar{\boldsymbol{x}}$.

Recap: RANSAC

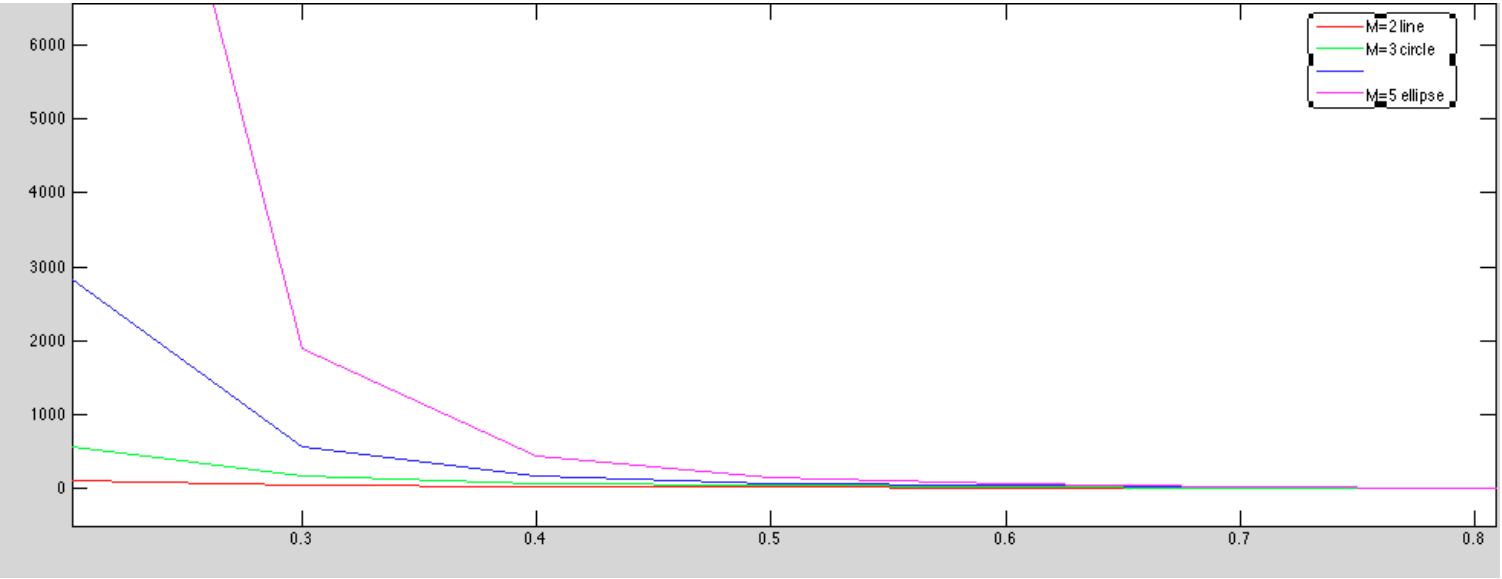


RANSAC = random hypotheses + a hard inlier test.

Calculating steps

$$N \log(1 - w^s) \leq \log(1 - p) \quad \Longleftrightarrow \quad N \geq \frac{\log(1-p)}{\log(1-w^s)}.$$

N is the number of trials before reaching a probability of p



About Frobenius Norm

For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the Frobenius norm is

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij}^2} = \sqrt{\mathrm{tr}(\mathbf{A}^\top \mathbf{A})} = \|\mathrm{vec}(\mathbf{A})\|_2.$$

Frobenius norm doesn't change after pre-multiplying by an orthonormal vector

Let $\mathbf{R} \in \mathbb{R}^{m \times m}$ be a rotation/orthogonal matrix, so $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$.

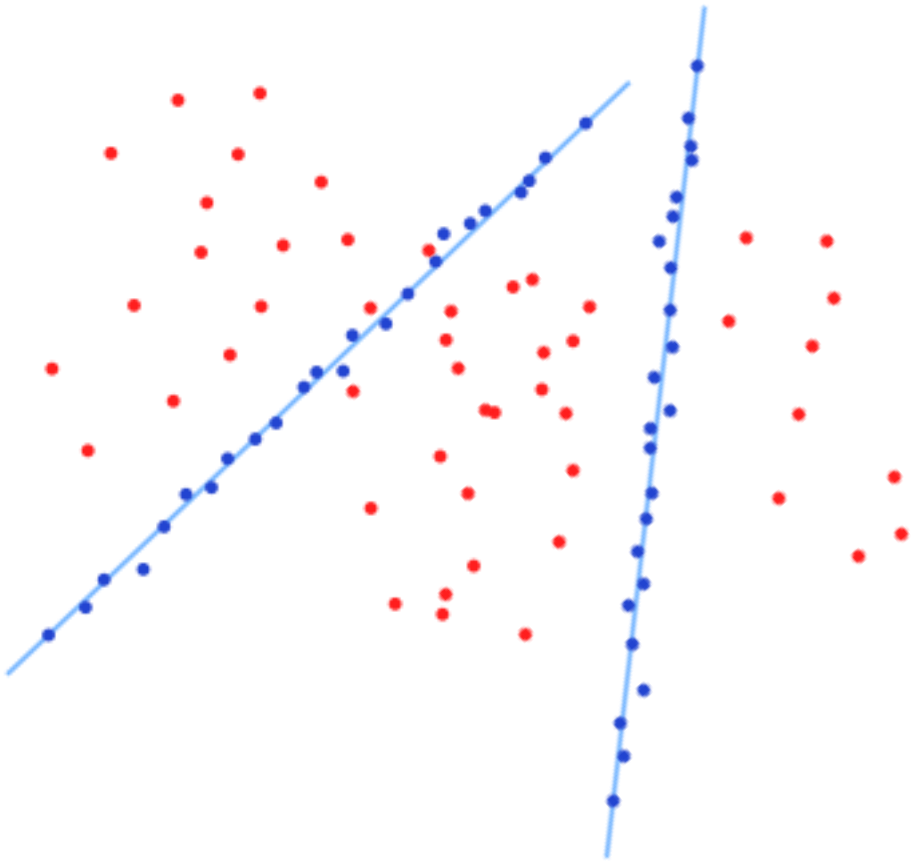
Compute using the trace form

$$\|\mathbf{RA}\|_F^2 = \mathrm{tr}\big((\mathbf{RA})^\top (\mathbf{RA})\big) = \mathrm{tr}\big(\mathbf{A}^\top \mathbf{R}^\top \mathbf{RA}\big) = \mathrm{tr}(\mathbf{A}^\top \mathbf{A}) = \|\mathbf{A}\|_F^2.$$

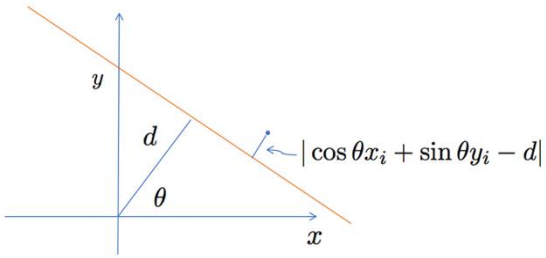
Rotating each column does not change the norm

Hough Transform

Multiple Lines



Norm form of a line



Normal form

A line is represented by (θ, ρ):

$$\rho = x\cos\theta + y\sin\theta,$$

where θ is the angle of the normal and ρ is distance from the origin.

Typical ranges: $\theta \in [0, \pi)$ and $\rho \in [-\rho_{\max}, \rho_{\max}]$.

In a Hough transform, we care about having as few parameters as possible

Why not homogeneous? We'll come back to this

Point in space = curve in parameters

For a fixed image point (x_0, y_0), all lines through it satisfy

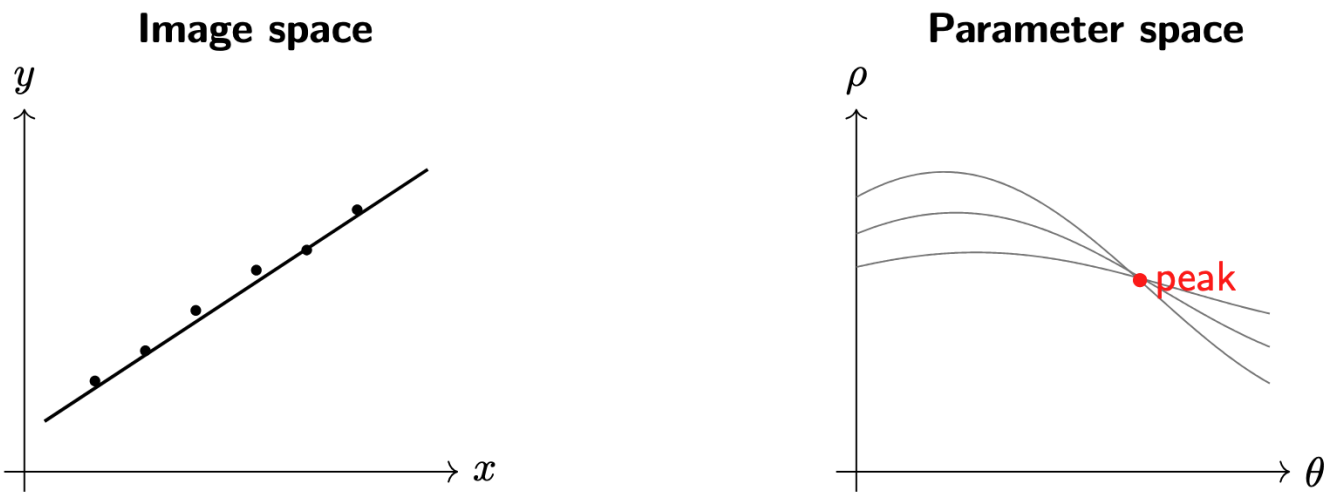
$$\rho(\theta) = x_0 \cos\theta + y_0 \sin\theta.$$

That is a sinusoid in (θ, ρ) space.

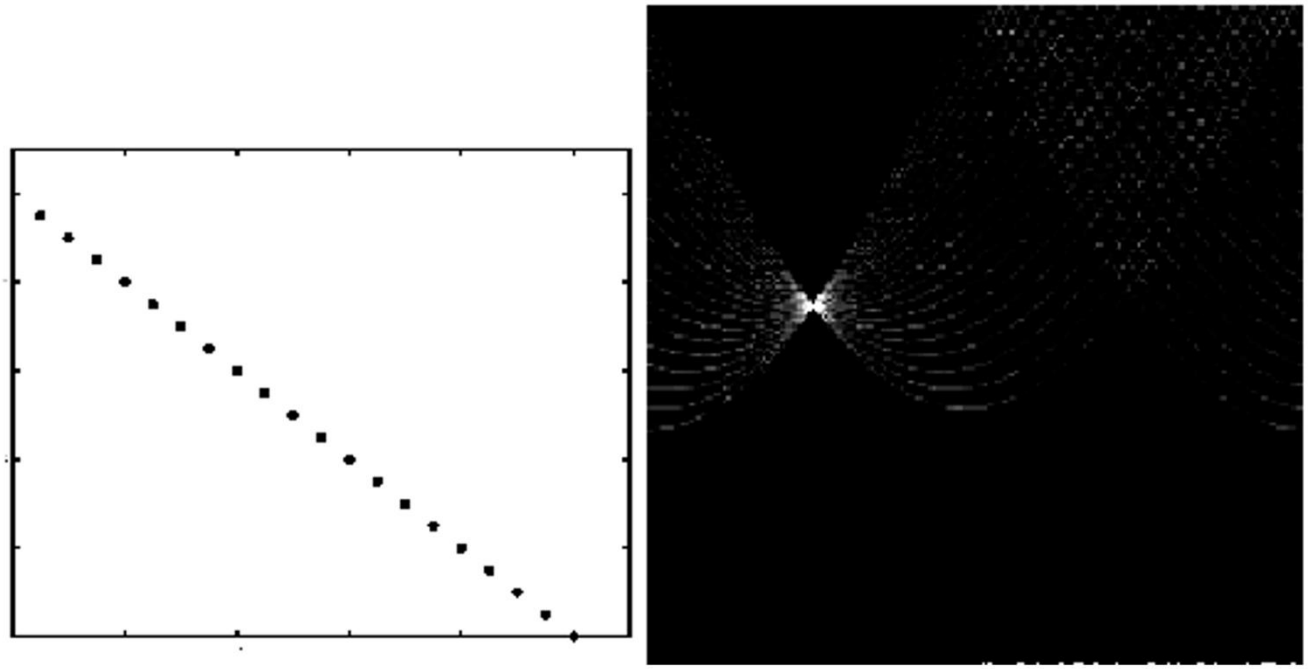
In practice:

```
import numpy as np
x0,y0 = 120.,80.
th = np.linspace(0, np.pi, 180, endpoint=False); rh = np.linspace(-300, 300, 601)
TH, RH = np.meshgrid(th, rh)
curve = x0*np.cos(TH) + y0*np.sin(TH)
mask = np.isclose(RH, curve, atol=(rh[1]-rh[0])/2)
rho_idx, theta_idx = np.where(mask)
```

Finding the best line



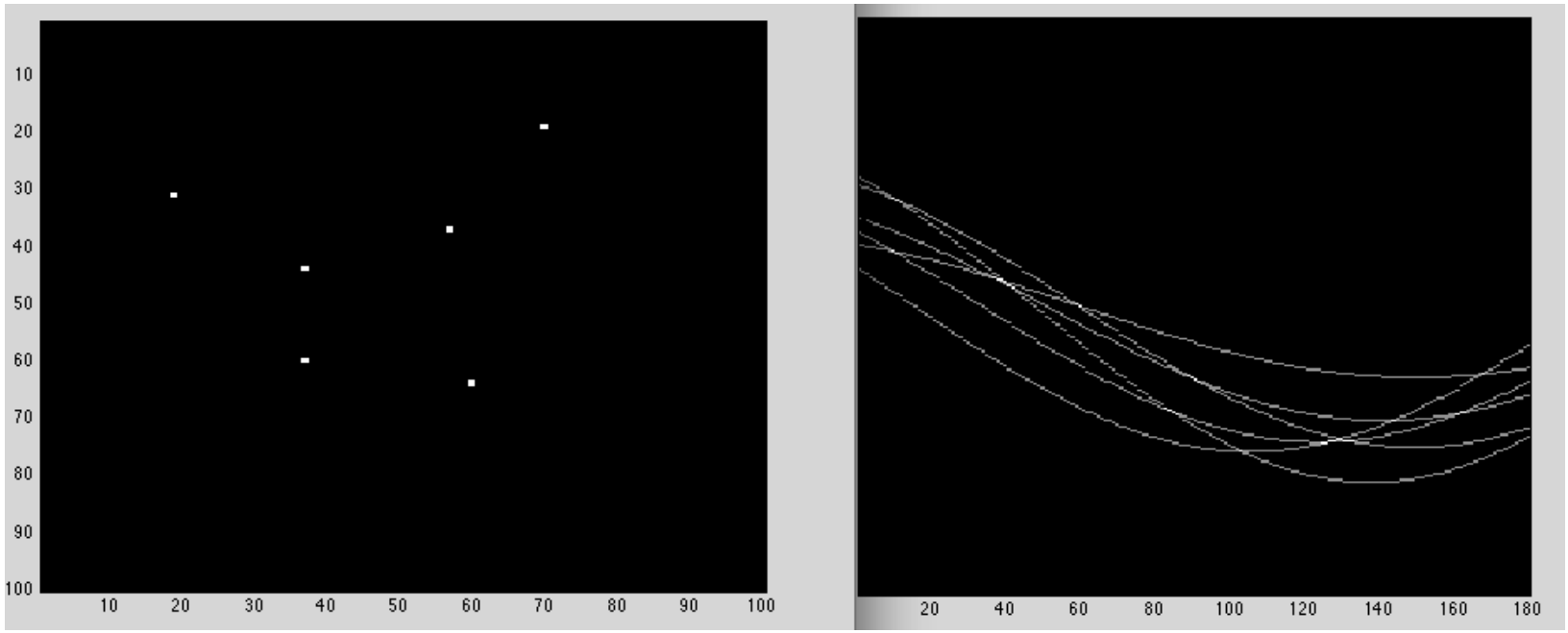
Example line



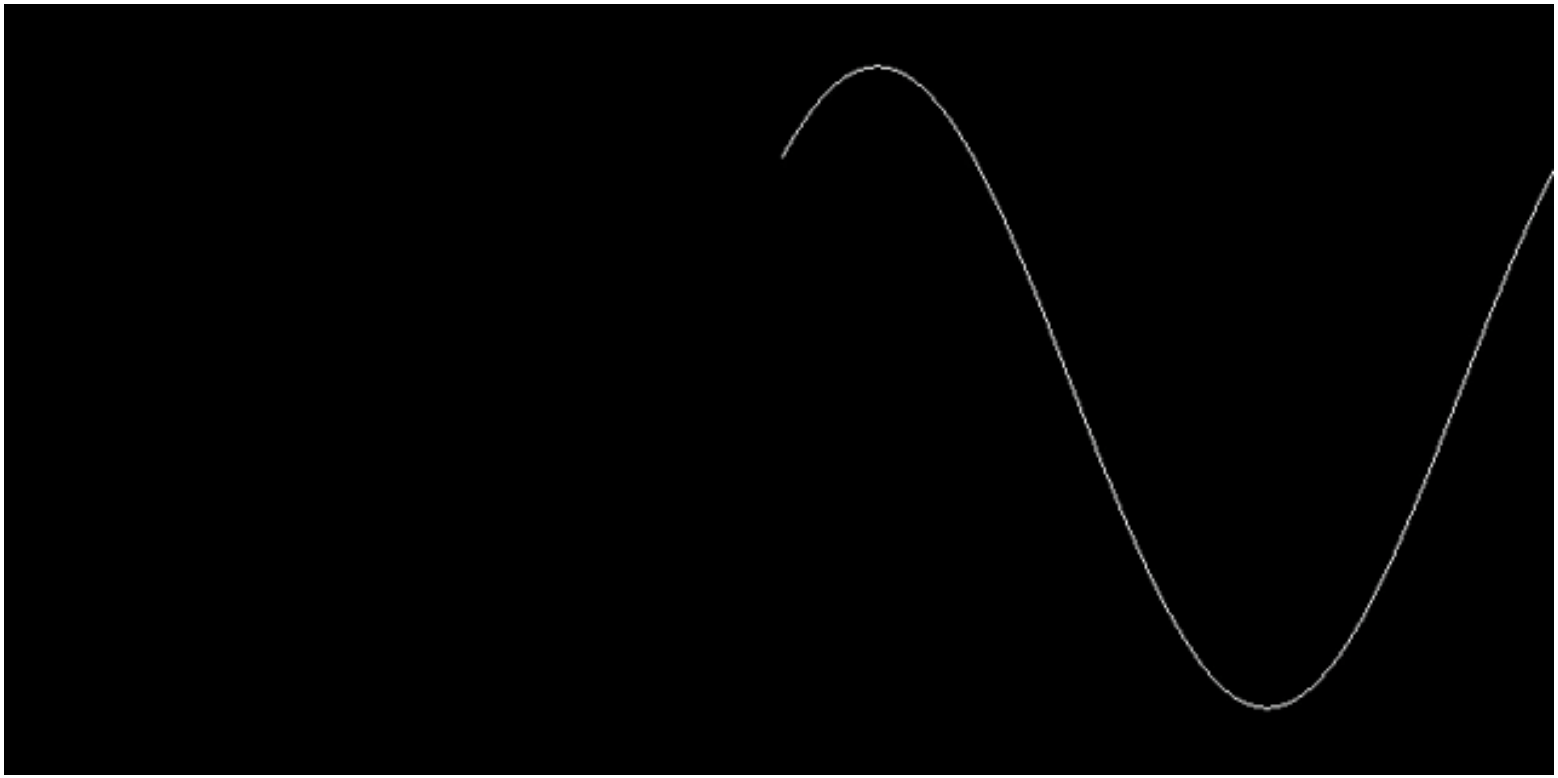
features votes

https://www.cs.rice.edu/~vo9/vision/slides/lecture08-RANSAC.pdf

Voting in the parameter space

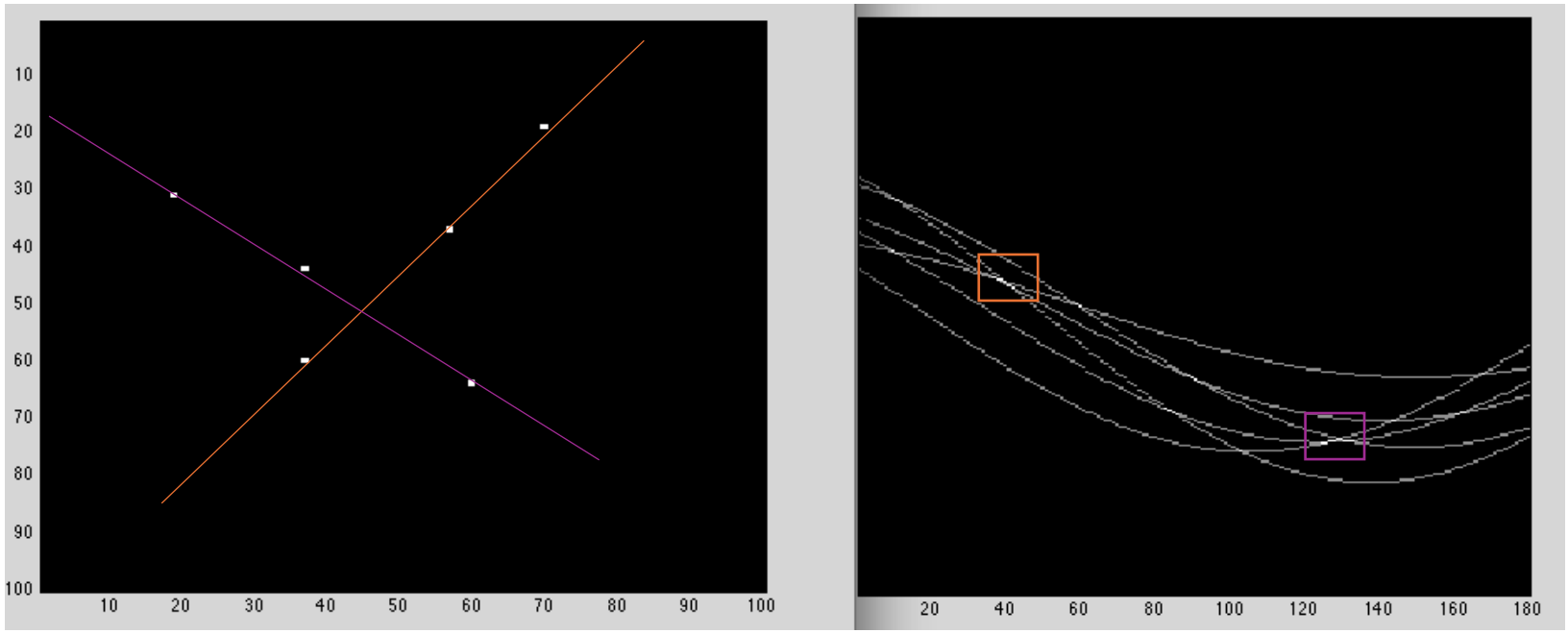


Example in motion



https://www.youtube.com/watch?v=ebfi7qOFLuo

Voting in the parameter space



Discretizing the parameters

We discretize the parameter space into bins:

- θ bins: e.g., 180 bins for 1° resolution "Curse of dimensionality"
- ρ bins: depends on image size and desired precision

Do we know why we avoid a homogeneous form of a line?

Voting

For each edge pixel (x, y) and each discrete θ_j:

$$\rho_j \leftarrow x \cos\theta_j + y \sin\theta_j \quad \Rightarrow \quad A[\theta_j, \rho_j] + +.$$

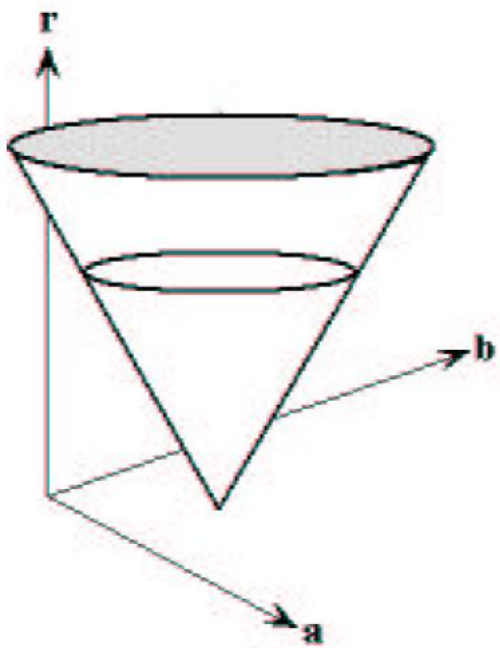
Then find peaks in A.

Example: circle detection

$$(x-a)^2 + (y-b)^2 = r$$

A point on the circle becomes a cone in (a, b, r)

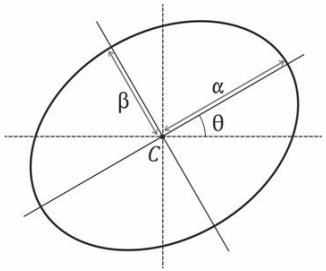
For each point, we put a ”vote” cone into the parameter space



Ellipsoid

Let us look at the ellipse equation

$$\begin{pmatrix} x - c_x & y - c_y \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \frac{1}{\alpha^2} & 0 \\ 0 & \frac{1}{\beta^2} \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x - c_x \\ y - c_y \end{pmatrix} = 1$$



Hough transform needs the computation of $V(c_x, c_y, \alpha, \beta, \theta)$ and search for maxima in 5-dimensional space. Hough becomes intractable beyond lines and circles.

Conversion to parameter space

- Assume the given point is fixed, and the model's parameters are variables

Line: $$\rho(\theta) = x_0 \cos\theta + y_0 \sin\theta.$$

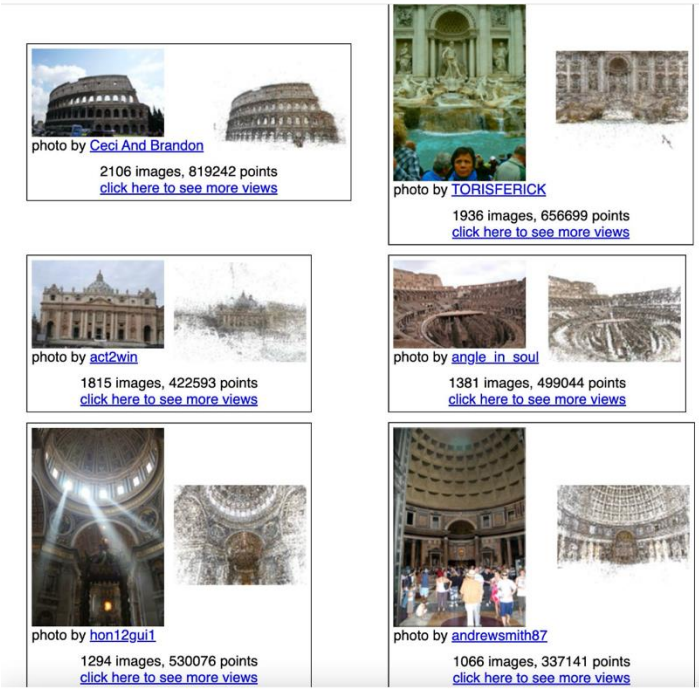
Sphere $$(x-a)^2 + (y-b)^2 = r$$

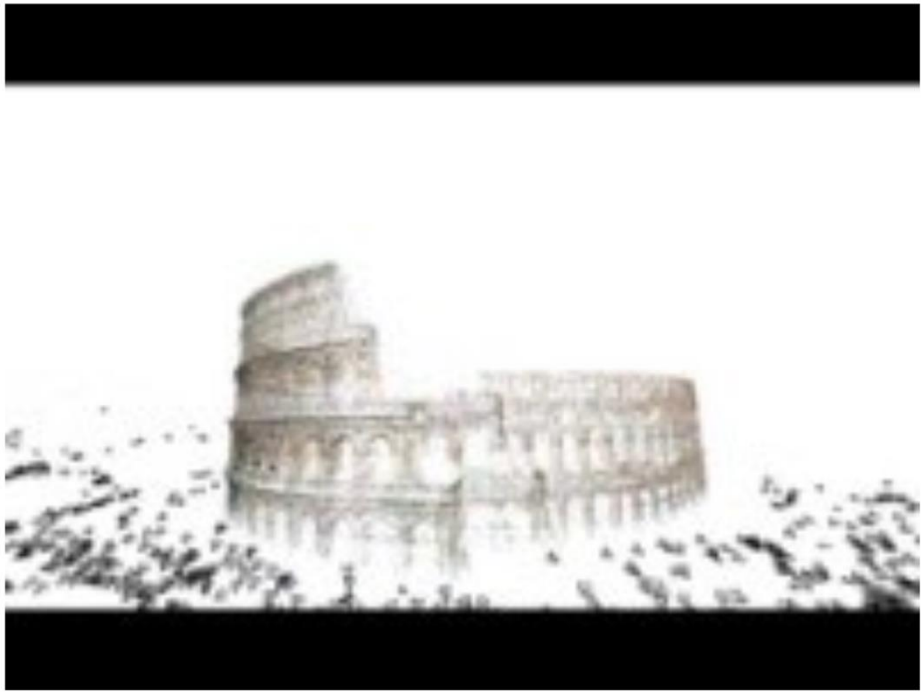
Summary

- RANSAC finds inlier set to estimate a single model
- Hough Transform detects multiple patterns
- We almost <u>never</u> directly estimate model parameters with single shot without robust filtering or outlier rejection

Two-view
Structure from Motion (SfM)

SfM Application: Building Rome in a Day





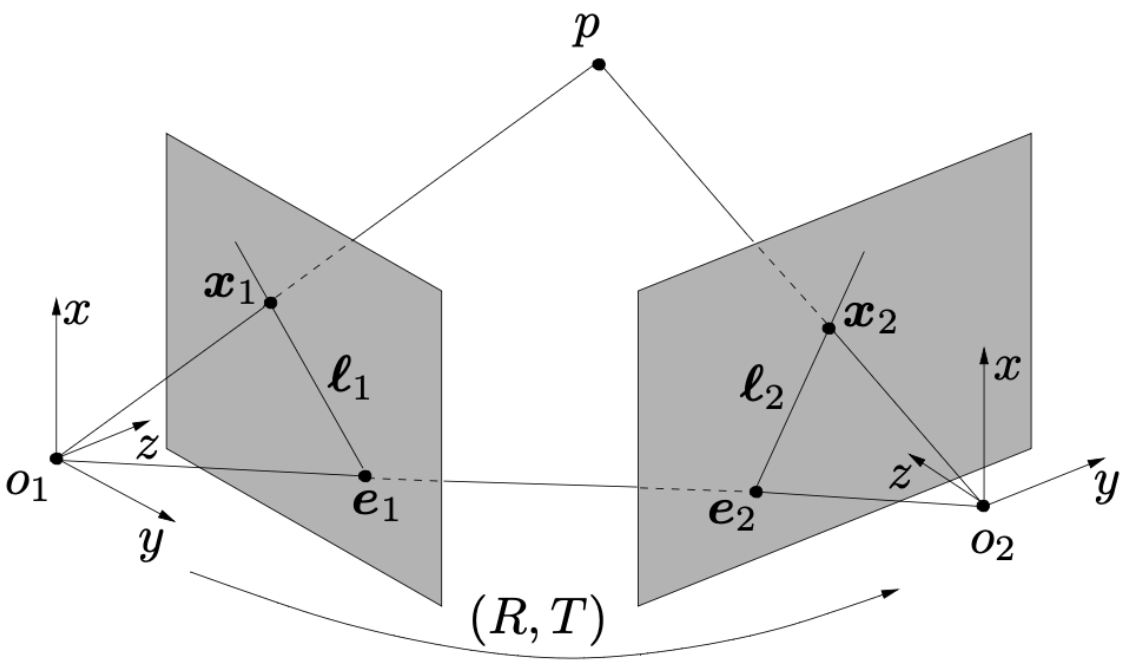
Meaning of two views

- A moving camera and a static scene
- Two cameras at different locations at the same time

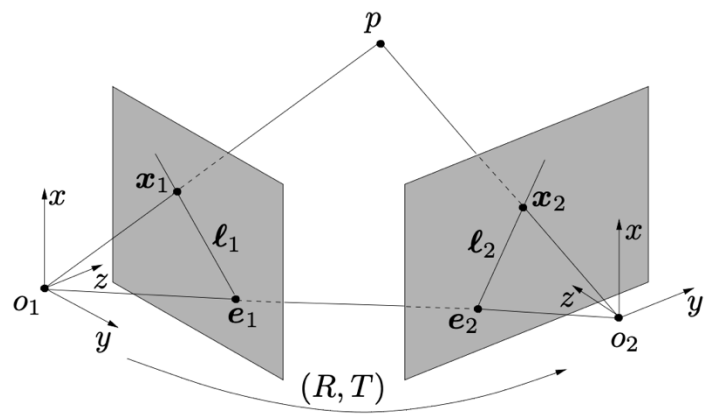
"We see because we move; we move because we see."
– James J. Gibson, *the Perception of the Visual World*

Calibrated two views

Two views


p
$\boldsymbol{x}_1$
$\boldsymbol{x}_2$
x
$\boldsymbol{\ell}_1$
$\boldsymbol{\ell}_2$
x
z
z
o_1
$\boldsymbol{e}_1$
$\boldsymbol{e}_2$
y
y
o_2
(R, T)

Epipolar constraint



Use normalized coordinates: $\boldsymbol{x} = (u, v, 1)^\top$.

Projection model

Camera 1: $\boldsymbol{x}_1 \sim \boldsymbol{X}$ (in camera-1 coordinates).
Camera 2: $\boldsymbol{x}_2 \sim \mathbf{R}\boldsymbol{X} + \mathbf{T}$.

Unknown depths appear as scale factors:

$$\lambda_1 \boldsymbol{x}_1 = \boldsymbol{X}, \qquad \lambda_2 \boldsymbol{x}_2 = \mathbf{R}\boldsymbol{X} + \mathbf{T}.$$

$$\boxed{\lambda_2 \boldsymbol{x}_2 - \mathbf{T} = \lambda_1 \mathbf{R} \boldsymbol{x}_1.}$$

Epipolar constraint

$$[\mathbf{T}]_\times = \begin{bmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{bmatrix}$$

Skew-symmetric matrix

$$\mathbf{T} \times (\lambda_2 \boldsymbol{x}_2 - \mathbf{T}) = \mathbf{T} \times (\lambda_1 \mathbf{R} \boldsymbol{x}_1).$$

The cross product of a vector with itself is zero:

$$\mathbf{T} \times \mathbf{T} = \mathbf{0}.$$

$$\lambda_2 (\mathbf{T} \times \boldsymbol{x}_2) = \lambda_1 (\mathbf{T} \times (\mathbf{R} \boldsymbol{x}_1)).$$

Epipolar Constraint

Dot with $\boldsymbol{x}_2^\top$:

$$\lambda_2 \, \boldsymbol{x}_2^\top (\mathbf{T} \times \boldsymbol{x}_2) = \lambda_1 \, \boldsymbol{x}_2^\top (\mathbf{T} \times (\mathbf{R}\boldsymbol{x}_1)).$$

But $\boldsymbol{x}_2^\top (\mathbf{T} \times \boldsymbol{x}_2) = 0$ because $(\mathbf{T} \times \boldsymbol{x}_2)$ is perpendicular to $\boldsymbol{x}_2$.

Therefore

$$\boldsymbol{x}_2^\top (\mathbf{T} \times (\mathbf{R}\boldsymbol{x}_1)) = 0,$$

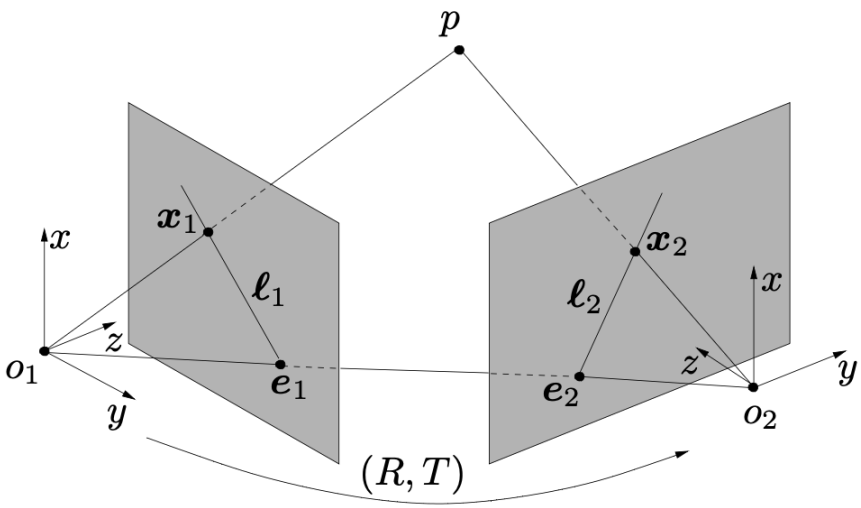
We got right of the scales by this series of algebraic manipulation

$$\boxed{\boldsymbol{x}_2^T \widehat{T} R \boldsymbol{x}_1 = 0}$$

What is the geometric intuition?

In the same coordinate system, three things lie on the same plane:

1. Ray of p in the second view (x2)
2. Ray of p in the second view, written in the first frame (R x2)
3. Vector O2 to O1, written in the first frame (T)



$$\boxed{\boldsymbol{x}_2^T \widehat{T} R \boldsymbol{x}_1 = 0}$$

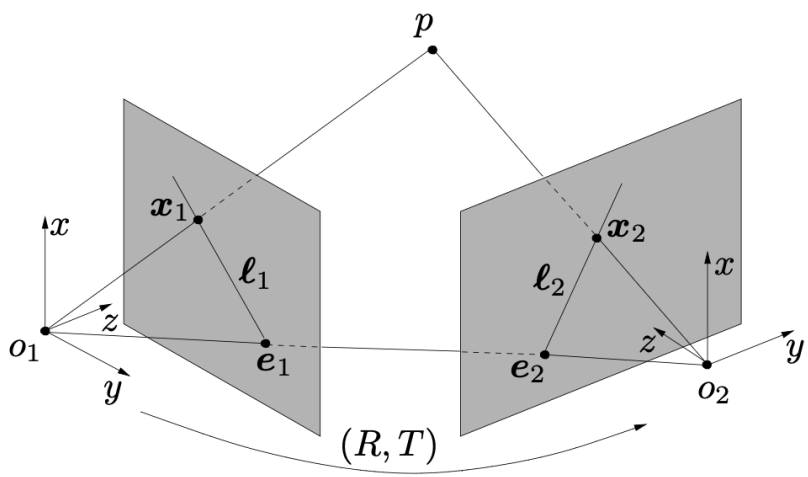
Epipolar Plane

Normal to epipolar plane

Define

$$\boldsymbol{n} \triangleq \boxed{\mathbf{T} \times (\mathbf{R}\boldsymbol{x}_1).}$$

Then $\boldsymbol{n} \perp \mathbf{T}$ and $\boldsymbol{n} \perp (\mathbf{R}\boldsymbol{x}_1)$, so $\boldsymbol{n}$ is a normal of the plane spanned by $\{\mathbf{T}, \mathbf{R}\boldsymbol{x}_1\}$.



$$\boxed{\boldsymbol{x}_2^T \widehat{T} R \boldsymbol{x}_1 = 0}$$

The bearing x_2 lies on the epipolar plane

Essential Matrix

Define the **Essential matrix**

$$\mathbf{E} \triangleq [\mathbf{T}]_\times \mathbf{R} \in \mathbb{R}^{3\times 3}.$$

Then the epipolar constraint is

$$\boxed{\boldsymbol{x}_2^\top \mathbf{E} \boldsymbol{x}_1 = 0.}$$

Properties of essential matrices

- E always has a rank of 2

Rank argument using full-rank $\mathbf{R}$

$\mathbf{R} \in SO(3)$ is invertible (full rank). Multiplying by an invertible matrix does not change rank:

$$\operatorname{rank}(\mathbf{E}) = \operatorname{rank}([\mathbf{T}]_\times \mathbf{R}) = \operatorname{rank}([\mathbf{T}]_\times).$$

Rank is the dimensionality of the space that all possible outputs span

For any vector $\boldsymbol{v}$,

$$[\mathbf{T}]_\times \boldsymbol{v} = \mathbf{T} \times \boldsymbol{v}.$$

$$\operatorname{rank}(\mathbf{E}) = \operatorname{rank}([\mathbf{T}]_\times) = 2.$$

The output vectors all live on a plane, normal to T

Properties of essential matrices

- E has <u>two identical singular values</u>

Let E be essential, so $E = [t]_\times R$ with $R \in SO(3)$ and $[t]_\times x = t \times x$.

- **Rotation does not change singular values.** Rotation does not change how much the matrix stretches space, so E and $[t]_\times$ have the same singular values.
- **Geometry of the cross product operator.** Decompose any $x \in \mathbb{R}^3$ as $x = x_\parallel + x_\perp$ with $x_\parallel \parallel t$ and $x_\perp \perp t$. Then

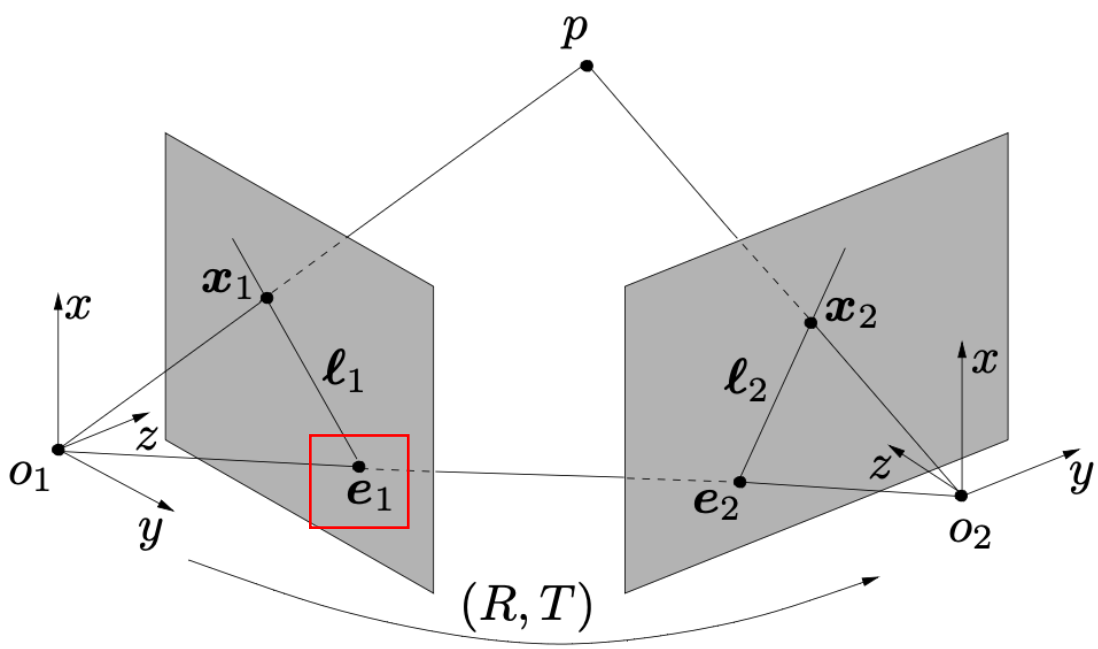
$$[t]_\times x_\parallel = t \times x_\parallel = 0 \quad \Rightarrow \quad \text{one singular value is } 0,$$

$$\|[t]_\times x_\perp\| = \|t \times x_\perp\| = \boxed{\|t\|\,\|x_\perp\| \quad \Rightarrow \quad \text{uniform scaling by } \|t\| \text{ on the 2D plane } t^\perp.}$$

No matter which direction, the vector is scaled by the same value $||t||$

Epipole

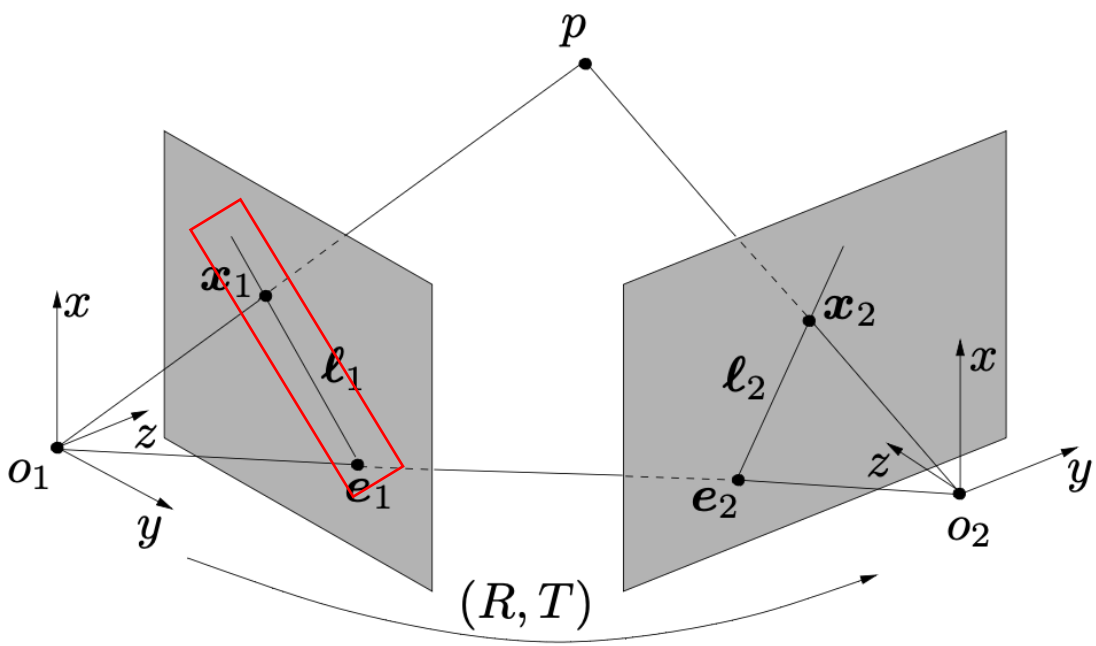
"The projection of the other camera's origin in the current image"



Also projection of the the direction of translation

Epipolar Lines

"Line connecting a pixel to the Epipole"



Points become lines in another view

The essential matrix maps a point in view 1 to an "epipolar line" in view 2

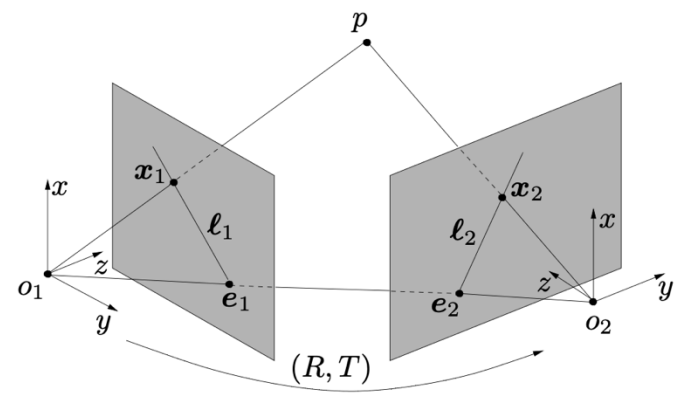
Given a point $\boldsymbol{x}_1$ in image 1, the corresponding epipolar line in image 2 is

$$\boldsymbol{\ell}_2 \sim \mathbf{E}\boldsymbol{x}_1.$$

Given $\boldsymbol{x}_2$ in image 2, the corresponding epipolar line in image 1 is

$$\boldsymbol{\ell}_1 \sim \mathbf{E}^\top \boldsymbol{x}_2.$$

A point becomes an epipolar line in second view



Start from the epipolar constraint

$$\boldsymbol{x}_2^\top \mathbf{E} \boldsymbol{x}_1 = 0.$$

Define

$$\boldsymbol{\ell}_2 := \mathbf{E}\boldsymbol{x}_1 \in \mathbb{P}^2.$$

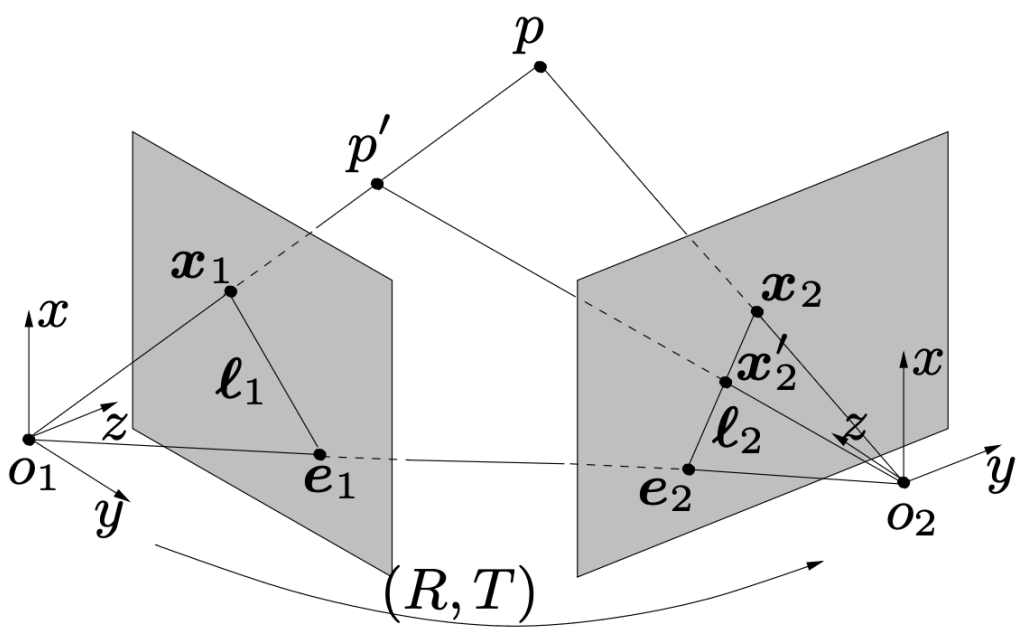
Then the constraint becomes the homogeneous line incidence equation

$$\boldsymbol{x}_2^\top \boldsymbol{\ell}_2 = 0,$$

so $\boldsymbol{\ell}_2$ is the epipolar line in image 2 on which the match $\boldsymbol{x}_2$ must lie.

Epipolar Lines

” An epipolar line is the line in the second image where the matching point must lie, because the 3D point lies somewhere along the first camera’s viewing ray.



Why do we care about epipolar lines

- If we know the pixel location in one view, and we know the essential matrix, the epipolar line allows us to search for the same point along a line instead of the whole image.
- **Important:** an essential matrix does not give us <u>point-to-point</u> correspondence.

Can we estimate R and T from E?

- Yes, but up to a scale in translation

SVD again!

Closest Essential matrix in Frobenius norm

$$\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top, \qquad \boxed{\sigma = \frac{\lambda_1 + \lambda_2}{2}.}$$

Because two identical singular values can also be set to all 1

Let $\mathbf{E} = \mathbf{U}\,\mathrm{diag}(\sigma, \sigma, 0)\,\mathbf{V}^\top$.

Then a translation direction is (up to sign)

$$\hat{\mathbf{T}} \sim \mathbf{U}(:, 3).$$